

Cyber Security

PDF 1 of the Series: Introduction, Types of Cyber Security, Security Teams and Prerequisites

Day-01 Class Notes: Fully Explained in Detail

What this PDF covers	Where it comes from in your notes
1. What is Cyber Security, Hacking and Information Security	Day-01, first page (definitions)
2. Digital assets, threats, virus and malware	Day-01, first page (side notes)
3. The 14 Types of Cyber Security	Day-01 last lines + continuation page
4. Information Security and GRC, OT-ICS Security	Continuation page
5. Red Team, Blue Team, Purple Team	Ethical hacking teams page
6. Prerequisites to become a Cyber Security Engineer	Same page, points 1 to 11
7. CIA Triad (Confidentiality, Integrity, Availability)	Prerequisite point 5
8. Revision summary and practice questions	Added for exam preparation

Next PDFs in this series: 2) Operating Systems and Servers, 3) Networking, OSI Model and IP Addressing, 4) Subnetting, Routing and Ports, 5) Remote Connectivity (RDP, SSH, WinRM), 6) Cloud Basics, 7) Linux Overview and Shells.

1. What is Cyber Security?

Your notes begin with the question "**What is cyber security?**" and give three related words: **cyber security**, **hacking** and **information security**. You also wrote that cyber security is "**a protection layer for digital assets**", and that digital assets can be things like a CCTV, scanner, printer, network asset, and any physical, digital or software-based asset. Let us understand each idea slowly.

1.1 The simple definition

Cyber security is the practice of protecting computers, servers, mobile phones, networks, applications, data and all connected electronic devices from **attack, damage, theft or unauthorized access**. The word *cyber* means anything related to computers, networks and the internet. The word *security* means protection. So cyber security literally means *protection of the digital world*.

Your note calls it a **protection layer**. This is a very good way to think about it. Imagine your house. You do not depend on only one lock. You have a compound wall, a gate, a door lock, an alarm and maybe a guard. Each of these is a *layer*. If a thief breaks one layer, the next layer still protects you. In the same way, cyber security is not one tool. It is many layers placed one over another around your digital assets. This idea is called **Defense in Depth**, and it explains why your notes list so many types of cyber security (physical, perimeter, application, cloud, and so on).

1.2 What is a digital asset?

An **asset** is anything of value to a person or an organization. A **digital asset** is any such thing that exists in or is connected to the digital world. In your notes you mention that assets can be **physical, digital and software-based**. Examples:

Asset type	Examples	Why it needs protection
Physical assets	CCTV camera, scanner, printer, server machine, laptop, router, switch, biometric machine	If someone steals or tampers with the device, they can reach the data inside it or use it to enter the network.
Digital assets	Databases, customer records, emails, documents, passwords, source code, backups, digital photos and videos	This is the actual information. Leaking or losing it causes money loss and loss of trust.
Software-based assets	Operating systems, web applications, mobile apps, licensed software, APIs	Bugs and misconfigurations in software are the most common way attackers get in.
Network assets	Firewalls, Wi-Fi, VPN, DNS, load balancers, internet links	The network is the road on which all data travels. Breaking it stops the business.

Notice that a CCTV, a printer or a scanner is also a computer today. It has an operating system, an IP address and a network connection. This is why attackers now target these "boring" devices: they are often forgotten and never updated.

1.3 Cyber Security vs Information Security vs Hacking

Your notes list these three together because students often mix them up. They are related but not identical.

Term	Meaning	Simple example
Cyber Security	Protecting everything in cyberspace: systems, networks, applications, devices and the data on them, from digital attacks.	Installing a firewall, patching a server, enabling multi-factor authentication.
Information Security (InfoSec)	Protecting information in any form: digital, paper or spoken. Its focus is on the three goals: confidentiality, integrity, availability. It also includes policy, rules and management.	A company rule that confidential documents must be locked in a cabinet, plus encryption of the same documents on the computer.
Hacking	Gaining access to a system or network by finding and using weaknesses in it. It can be legal (ethical hacking, done with permission) or illegal (done to harm).	A security tester finding a weak password on a company server with written permission, versus a criminal doing the same to steal money.

So how do they relate? **Information security** is the wider umbrella of protecting information. **Cyber security** is the part of it that deals with the digital and internet-connected world. **Hacking** is the activity that either attacks these protections (bad) or tests them (good). In your notes you wrote that information security is also a "protection layer for digital assets" which matches this idea.

1.4 Types of hackers (important background)

- **White hat hacker (ethical hacker):** works with permission to find weaknesses and help fix them. This is the career path you are learning.
- **Black hat hacker:** breaks into systems illegally for money, revenge, fame or spying.
- **Grey hat hacker:** breaks in without permission but usually without a harmful goal, and often informs the owner afterwards. It is still illegal in most countries.
- **Script kiddie:** a beginner who uses ready-made tools without understanding them.
- **Hacktivist:** hacks to support a political or social cause.

Exam / Interview Tip: If asked "Is hacking illegal?", the correct answer is: *hacking without permission is illegal; hacking with written permission (ethical hacking or penetration testing) is a legal, well-paid profession.*

2. Threats: Attacks, Unauthorized Access, Virus and Malware

On the first page you also wrote words such as **attack**, **unauthorized access**, **virus**, **malware**. These are the enemies that cyber security fights. Let us define each carefully.

2.1 Attack and unauthorized access

An **attack** is any deliberate attempt to damage, steal from, disturb or take control of a digital asset. **Unauthorized access** means entering a system, account or file that you have no permission to use. For example, guessing someone's password and logging into their email is unauthorized access. Almost every big cyber attack starts with unauthorized access at some point.

2.2 Malware

Malware means *malicious software*, that is, any program written with the aim of causing harm. "Virus" in your notes is one kind of malware. The main kinds are:

Type	How it behaves
Virus	Attaches itself to a clean file or program and spreads when the infected file is opened or shared. Needs a human action to spread.
Worm	Spreads by itself through the network without any human action. It can infect thousands of machines in minutes.
Trojan	Looks like a useful program (for example a free game) but secretly does harm, such as opening a back door for the attacker.
Ransomware	Encrypts all your files and demands money to give them back.
Spyware / Keylogger	Secretly watches you and records what you type, including passwords.
Adware	Shows unwanted advertisements and sometimes tracks your browsing.
Rootkit	Hides deep inside the operating system so that even antivirus cannot easily see it.
Botnet	A large group of infected computers controlled by an attacker, used to launch big attacks such as DDoS.

2.3 Why do attacks happen?

- **Money:** ransom, selling stolen data, stealing bank details.
- **Espionage:** stealing company or government secrets.
- **Disruption:** stopping a service to hurt a competitor or country.
- **Revenge or ideology:** a former employee or an activist.
- **Challenge and fame:** proving skill.

2.4 Common attack vectors (ways attackers get in)

- Weak, reused or stolen passwords.
- Phishing emails and fake websites that trick people into giving information.
- Unpatched software with known vulnerabilities.
- Misconfigured servers, open ports, open cloud storage.
- Infected USB drives and downloads.
- Insider threats: a careless or dishonest employee.
- Physical access to a device.

Exam / Interview Tip: Remember the difference: **Vulnerability** = a weakness. **Threat** = something that can exploit the weakness. **Risk** = the chance and impact of the threat exploiting the weakness. Interviewers ask this very often.

3. The Types of Cyber Security (from your notes)

At the bottom of the Day-01 page you wrote "**Types of cyber security: 1) Physical security 2) Perimeter security**", and on the next page the list continues from 3 to 14: application, IAM, cloud, data, e-mail, AI, OS, hardware, firmware, embedded, container and network security. Together these make **14 types**. Each type protects a different layer, so a company needs all of them, not just one. Below, each one is explained with what it protects, how it is done, and a real example.

No.	Type	One-line meaning
1	Physical Security	Protecting buildings, rooms and devices from physical harm or theft
2	Perimeter Security	Protecting the outer boundary of the network
3	Application Security	Making software and websites free of exploitable bugs
4	IAM Security	Controlling who can access what (identity and access management)
5	Cloud Security	Protecting resources kept on cloud platforms
6	Data Security	Protecting the data itself, wherever it goes
7	E-mail Security	Protecting mailboxes from phishing, spam and malware
8	AI Security	Protecting AI systems and preventing misuse of AI
9	OS Security	Hardening Windows, Linux, macOS and other operating systems
10	Hardware Security	Protecting the chips and physical components
11	Firmware Security	Protecting the low-level software that runs on hardware
12	Embedded Security	Protecting small dedicated computers inside devices
13	Container Security	Protecting Docker/Kubernetes style containers
14	Network Security	Protecting data moving across networks

3.1 Physical Security

Physical security means stopping unauthorized people from physically touching your equipment. The best software security is useless if a thief can walk into the server room and take the hard disk. Methods include security guards, fences, ID cards and biometric locks, CCTV cameras, locked server racks, visitor registers, fire alarms and sprinklers, and power backup. Your notes mention CCTV and scanners as assets; note that CCTV is both a *tool* of physical security and an *asset* that must itself be protected from hacking.

3.2 Perimeter Security

The **perimeter** is the boundary between your trusted internal network and the untrusted outside world (the internet). Perimeter security is like the main gate and compound wall of a company. Typical tools

are **firewalls** (allow or block traffic using rules), **IDS/IPS** (Intrusion Detection and Prevention Systems which watch for attack patterns), **DMZ** (a separate zone for public-facing servers so that a hacked web server does not give direct access to the inside), **VPN gateways**, **proxy servers** and **web application firewalls (WAF)**. Modern companies also say that the perimeter alone is not enough because attackers or malicious insiders may already be inside; that is why other layers exist.

3.3 Application Security

Application security (AppSec) is about building and testing software so that attackers cannot misuse it. Web and mobile apps are a favourite target because they are open to everyone on the internet. Typical problems are described by the **OWASP Top 10**, for example SQL injection, cross-site scripting (XSS), broken authentication, insecure direct object references and security misconfiguration. Protective practices include secure coding, input validation, code review, SAST/DAST scanning, penetration testing, patching libraries and using a WAF. The idea of *DevSecOps*, adding security at every stage of software development, belongs here.

3.4 IAM Security (Identity and Access Management)

Your note writes it as "**IAM security (Identity access management)**". IAM answers three questions: **Who are you?** (authentication), **What are you allowed to do?** (authorization), and **What did you do?** (accounting or auditing). Key ideas are strong passwords, **multi-factor authentication (MFA)**, single sign-on (SSO), role-based access control (RBAC), and the **principle of least privilege**, which says every user should get only the minimum access needed for the job. IAM is also a core service in cloud platforms such as AWS, and your Day-02 cloud notes mention IAM in the list of cloud domains.

3.5 Cloud Security

Cloud security protects data, applications and infrastructure hosted on platforms like AWS, Microsoft Azure and Google Cloud. A key concept is the **Shared Responsibility Model**: the cloud provider secures the physical data centers and the underlying infrastructure, but *you* are responsible for how you configure your servers, storage, network rules, identities and data. Most cloud breaches happen because of customer mistakes such as a publicly open storage bucket or an over-permissive IAM role, not because the provider was hacked. Common tools: security groups, encryption at rest and in transit, key management, logging and monitoring, and Cloud Security Posture Management (CSPM).

3.6 Data Security

Data security protects the information itself, whether it is stored (at rest), moving over a network (in transit) or being processed (in use). Techniques include **encryption**, hashing, tokenization and masking, access control, backup and recovery, and **Data Loss Prevention (DLP)** tools that stop sensitive information from leaving the company. Laws such as GDPR in Europe and the DPDP Act in India make organizations legally responsible for protecting personal data.

3.7 E-mail Security

Email is the number one entry point for attackers. E-mail security defends against **phishing** (fake emails that trick a user), spear-phishing (targeted phishing), business email compromise, spam and malicious attachments. Defenses include spam filters, attachment sandboxing, link scanning, and the DNS-based authentication standards **SPF**, **DKIM** and **DMARC** which prove that an email really came

from the claimed domain. User awareness training is equally important because the last line of defense is the human who clicks.

3.8 AI Security

In your notes this appears as "AI security". It has two sides. First, **protecting AI systems** from attacks such as prompt injection, data poisoning (feeding bad training data), model theft and adversarial inputs. Second, **securing the use of AI** in a company, for example stopping employees from pasting confidential data into public AI tools and checking AI-generated code for vulnerabilities. It is a fast-growing area of the field.

3.9 OS Security

The operating system controls everything on a computer, so hacking the OS gives full control. OS security includes regular patching, removing unnecessary services, strong user and permission management, enabling the built-in firewall, using antivirus/EDR, disk encryption (BitLocker on Windows, LUKS on Linux) and following **hardening** guides such as the CIS Benchmarks. Since your course next covers OS overview and Linux, this topic will be studied in depth in PDF 2 and PDF 7.

3.10 Hardware Security

Hardware security protects the physical components of a computer such as the CPU, memory, motherboard and storage. It includes **TPM chips** (Trusted Platform Module) that safely store encryption keys, **Secure Boot**, hardware security modules (HSMs), and defence against side-channel attacks and hardware backdoors placed during manufacturing. It also covers disabling unused ports such as USB.

3.11 Firmware Security

Firmware is the small permanent software stored on a hardware chip that starts and controls it, for example the BIOS/UEFI of a computer, or the software inside a router or printer. Because it runs below the operating system, malware in firmware survives reinstalling the OS and is very hard to detect. Protection involves signed firmware updates, Secure Boot, timely updates from the vendor and integrity checks.

3.12 Embedded Security

An **embedded system** is a small computer built into a device to do one job: a smart TV, ATM, car ECU, medical device, smart meter, CCTV camera or IoT sensor. These often have weak default passwords, old software and no easy way to patch. Embedded security means changing default credentials, secure design, encrypted communication, signed updates and limiting network exposure. It ties in with your "CCTV" asset example.

3.13 Container Security

Containers (Docker, Kubernetes) package an application with everything it needs so it runs the same everywhere. Because many containers share one host kernel, one insecure container can put others at risk. Container security covers scanning images for vulnerabilities before use, using trusted base images, not running containers as root, limiting resources and permissions, protecting secrets, and securing the orchestration layer (Kubernetes API, network policies).

3.14 Network Security

Network security protects the network infrastructure and the data moving over it. Typical controls are firewalls, IDS/IPS, network segmentation (VLANs and subnets), VPN, secure Wi-Fi (WPA3), NAC (Network Access Control), DDoS protection and monitoring with a SIEM. It relies heavily on knowing how networks work, which is why your notes then move into the OSI model, IP addresses and subnetting (covered in PDF 3 and PDF 4).

Exam / Interview Tip: If asked to "list the types of cyber security", write at least: network, application, cloud, data, IAM, endpoint/OS, physical and email security, and explain that they work together as layers.

4. Information Security and GRC

In your notes, point 2 on the continuation page is "**Information security: GRC (Under Governance, Risk and Compliance)**". This is the *management* side of security, as opposed to the technical side.

4.1 What GRC means

Letter	Meaning	Explanation and example
G	Governance	The rules, policies, roles and responsibilities of security in the organization. Who is in charge? What is the password policy? Example: The board approves an Information Security Policy and appoints a CISO.
R	Risk	Finding, measuring and reducing the risks to assets. Example: Doing a risk assessment and finding that unencrypted laptops are a high risk, then deciding to enforce disk encryption.
C	Compliance	Following laws, regulations and standards, and proving it with audits. Example: ISO 27001, PCI-DSS for card payments, HIPAA for health data, GDPR, SOC 2, India's DPDP Act.

4.2 Why GRC matters

Technology alone cannot make a company secure. Someone must decide *what* to protect, *how much* to spend, *who* is responsible and *whether* the company follows the law. GRC professionals write policies, run risk assessments, manage audits and coordinate with regulators. Many people who do not like deep technical work choose this career path in cyber security. Important related documents are: **Policy** (high-level rule), **Standard** (specific mandatory requirement), **Procedure** (step by step instructions) and **Guideline** (recommended advice).

4.3 Risk management basics

- **Identify assets** and decide their value.
- **Identify threats and vulnerabilities** that affect each asset.
- **Calculate risk** = likelihood x impact.
- **Treat the risk:** reduce it (add controls), avoid it (stop the activity), transfer it (buy insurance or outsource) or accept it (knowingly live with a low risk).
- **Monitor and review** continuously, because risks change.

5. OT-ICS Security

Your note reads "**OT - ICS Security (Operational Technology - Industrial Control System)**" with the example: *big industries, securing the industry under security, nothing but security; industries, PLC, SCADA, ISU and many manufacturing industries etc.*

5.1 IT versus OT

IT (Information Technology) is the world of office computers, servers, databases and networks that handle *information*. **OT (Operational Technology)** is the world of machines that control *physical*

processes: opening a valve, spinning a turbine, moving a conveyor belt or switching power on a grid. An **Industrial Control System (ICS)** is the collection of hardware and software that runs those physical processes.

Component	Meaning	Role
PLC	Programmable Logic Controller	A rugged industrial computer that reads sensors and controls machines by following a program.
SCADA	Supervisory Control and Data Acquisition	Software and networks that let operators monitor and control many remote sites, such as pipelines or power stations, from one control room.
HMI	Human Machine Interface	The screen on which operators see the process and give commands.
RTU	Remote Terminal Unit	A field device that collects data from remote sensors and sends it to SCADA.
DCS	Distributed Control System	A control system for large plants like refineries where control is spread across many controllers.

5.2 Why OT/ICS security is different and important

- In IT the top priority is **Confidentiality**. In OT the top priority is **Availability and Safety**, because a stopped plant or a wrong command can hurt people or the environment.
- OT systems often run for 20 to 30 years and use old operating systems that cannot be patched easily.
- They were originally built to be isolated, without security in mind. Now they are connected to IT networks and the internet, which exposes them.
- Real world example: the **Stuxnet** malware (discovered in 2010) damaged industrial centrifuges by attacking PLCs, and attacks on power grids in Ukraine caused blackouts. Such attacks prove that cyber attacks can cause physical damage.
- Industries that need OT security: power and energy, oil and gas, manufacturing, water treatment, railways, metro, pharmaceuticals, mining and smart buildings.

Exam / Interview Tip: Frameworks for OT security: **IEC 62443** and the **Purdue Model** (which divides the industrial network into levels). Mentioning them in an interview shows depth.

6. Ethical Hacking Teams: Red, Blue and Purple

The page titled "**Under cyber security, ethical hacking team**" lists three teams. They come from the idea of a war game or exercise in which one side attacks and the other side defends, so that the organization learns to become stronger.

Team	Your note says	Detailed meaning
Red Team	Offensive security	Acts like a real attacker. Tries to break into the organization using any allowed method: technical attacks, phishing, social engineering, even physical entry. Goal: find weaknesses <i>before</i> real criminals do. Typical roles: Penetration Tester, Ethical Hacker, Red Team Operator, Exploit Developer.
Blue Team	Defensive security	Protects the organization. Monitors systems 24 x 7, detects attacks, investigates alerts and responds. Typical roles: SOC Analyst, Incident Responder, Threat Hunter, Security Engineer, Digital Forensics Analyst. Tools: SIEM, EDR, firewall, IDS/IPS.
Purple Team	Collaboration between Red and Blue team	Not a separate group of people so much as a way of working. Red and Blue sit together: red shows the attack it used, blue checks whether it was detected, and together they improve detection rules and defenses immediately. This removes the "us versus them" problem.

6.1 Example scenario

Suppose the Red Team sends a phishing email to employees and steals one password. The Blue Team checks its logs: did the SIEM raise an alert for the strange login? If not, there is a gap. In a purple exercise both teams review it together and the Blue Team writes a new detection rule and blocks the technique. Next time the attack is caught in seconds.

6.2 Extra colours you may hear

- **Yellow team:** builders (software developers) who create the systems.
- **Green team:** bridges the blue team and the builders to improve security in development.
- **White team:** the referee that sets the rules of the exercise.
- **Orange team:** helps red team members share knowledge with the blue team.

The three you must know for now are Red, Blue and Purple.

6.3 Career paths

If you like...	Choose	Example job titles
Breaking into systems, creative thinking	Red Team (Offensive)	Penetration Tester, Red Team Engineer, Bug Bounty Hunter
Monitoring, investigation, defending	Blue Team (Defensive)	SOC Analyst L1/L2/L3, Incident Responder, Threat Hunter

If you like...	Choose	Example job titles
Improving both sides, detection engineering	Purple Team	Detection Engineer, Security Architect, Adversary Emulation Specialist
Policies, audits, rules	GRC	Risk Analyst, Compliance Auditor, ISO 27001 Lead Auditor

7. Prerequisites: To Become a Cyber Security Engineer

Your teacher wrote the heading **"To become a CS engineer, following:"** and then 11 skills. Cyber security is not a beginner's first subject; it stands on top of several IT foundations. Here is each one explained: what it is, why it is needed, and what you should learn.

No.	Prerequisite (from your notes)	Why you need it
1	OS knowledge	Attackers and defenders both work inside operating systems.
2	Networking	All attacks travel through networks.
3	Scripting or programming language	To automate tasks and understand exploits.
4	Basic virus and antivirus knowledge	To understand malware and defenses.
5	Basic cyber security knowledge (CIA)	The core principles.
6	Basic database knowledge	Data is the main target.
7	Basic DNS knowledge	The internet's address book.
8	Basic certificates	The basis of HTTPS and trust.
9	Basic CDN (Content Delivery Network) knowledge	Modern websites are served through CDNs.
10	Basic web server knowledge	Web servers are exposed to the internet.
11	Basic load balancer and auto scaling	Modern cloud application design.

7.1 Operating System knowledge

Learn how Windows, Linux and macOS work: files and folders, users and permissions, processes, services, memory, logs and the command line. Linux is especially important because most servers, cloud machines and security tools (like Kali Linux) use it. Your notes go into this topic next: types of OS (client or end-user OS such as Windows 10, 11, 8.1; server OS such as Windows Server 2025, 2022, 2019, 2016, 2012), OS overview and Linux families.

7.2 Networking

You must understand how two computers talk: the OSI model, IP addresses, classes, private and public IP, subnetting, ports, protocols, routing, TCP versus UDP, DNS and HTTP. Without networking you cannot understand scanning, sniffing, firewalls or most attacks. This is the subject of PDF 3 and

7.3 Scripting or Programming Language

Your note first wrote "scripting" and then "programming language". Scripting means writing small programs to automate tasks, for example a Bash script to check many servers or a Python script to scan ports. Recommended: **Python** (most popular in security), **Bash** for Linux, **PowerShell** for Windows, and later JavaScript and SQL for web attacks. You do not have to be a software developer, but you must be able to read code and write simple tools.

7.4 Basic virus and antivirus knowledge

Know what malware types exist (see section 2.2), how they spread, and how antivirus works: **signature-based detection** (matching known malware patterns), **heuristic and behaviour-based detection** (spotting suspicious behaviour), and modern **EDR** tools. Also know the limits: antivirus cannot catch everything, especially new (zero-day) malware.

7.5 Basic cyber security knowledge: the CIA Triad

Your note writes "[CIA, ...] **Confidentiality, Integrity**". This is the CIA Triad, the foundation of all security. It has nothing to do with the intelligence agency.

Principle	Meaning	Example of a failure	Example of protection
Confidentiality	Only authorized people can see the information.	A hacker reads customer credit card numbers.	Encryption, passwords, MFA, access control.
Integrity	Information stays accurate and is not changed without permission.	An attacker changes the amount in a bank transfer.	Hashing, digital signatures, checksums, version control, file permissions.
Availability	Systems and data are available when needed.	A DDoS attack takes a website down; ransomware locks files.	Backups, redundancy, load balancers, DDoS protection, disaster recovery.

Some notes on the page show the letters CIA followed by extra words. Beyond the triad, security also uses **Authentication** (proving identity), **Authorization** (permissions), and **Non-repudiation** (a person cannot deny they did an action, achieved using digital signatures and logs).

7.6 Basic database knowledge

Almost all valuable data lives in databases such as MySQL, PostgreSQL, Oracle, SQL Server and MongoDB. Learn basic SQL (SELECT, INSERT, UPDATE, DELETE, WHERE, JOIN) and how applications use databases. This lets you understand **SQL injection**, one of the most famous attacks, in which an attacker inserts SQL commands through an input field to read or change the database.

7.7 Basic DNS knowledge

DNS (Domain Name System) converts human-friendly names like *www.example.com* into IP addresses. It is like the phone book of the internet. Understand DNS records (A, AAAA, CNAME, MX,

NS, TXT), the resolution process (browser cache, resolver, root server, TLD server, authoritative server) and attacks like DNS spoofing/poisoning and DNS tunneling. DNS is also a big part of your assignment about what happens when you enter a URL.

7.8 Basic certificates

A **digital certificate** (SSL/TLS certificate) proves that a website really belongs to who it says it does and enables encrypted **HTTPS** connections. A trusted **Certificate Authority (CA)** signs it. Learn about public and private keys, the TLS handshake, certificate expiry, self-signed versus CA-signed certificates and what browser warnings such as "Your connection is not private" mean.

7.9 Basic CDN (Content Delivery Network) knowledge

A **CDN** is a global network of servers (like Cloudflare, Akamai, Amazon CloudFront) that store copies of a website's content close to users so pages load faster. A user in Bengaluru gets the content from a nearby server rather than one in America. CDNs also absorb DDoS attacks and hide the real server's IP address, so they are important for both performance and security.

7.10 Basic web server knowledge

A **web server** (Apache, Nginx, Microsoft IIS) receives HTTP/HTTPS requests from browsers and sends back web pages. Learn the request-response cycle, HTTP methods (GET, POST), status codes (200, 301, 403, 404, 500), headers, cookies and sessions, and common web server misconfigurations. Web servers are the first thing an attacker sees, so they must be hardened.

7.11 Basic load balancer and auto scaling

A **load balancer** spreads incoming traffic across multiple servers so that no single server is overloaded and, if one fails, others continue. **Auto scaling** automatically adds servers when traffic rises and removes them when it falls. Both are key in cloud environments and they help availability, the "A" of the CIA triad. Your note also underlines the word "Tradeoffs"/"Tracker" next to it, which is a reminder that these design choices should be understood, not just memorized.

8. Your Assignment: What Happens When You Enter a URL in the Browser?

Your notes give this as **Assignment 1**: "When you enter a URL in the browser, what happens? How traffic will flow." The full answer needs networking and DNS, so it will be built up over the next PDFs. Here is a preview outline so you can start preparing. This is also one of the most frequently asked interview questions.

- **Step 1 - URL parsing:** The browser splits the URL into protocol (https), domain (www.example.com), port (443 by default) and path.
- **Step 2 - Cache check:** The browser, the operating system and the hosts file are checked for a saved IP address.
- **Step 3 - DNS resolution:** If not found, a DNS query goes to the configured resolver, which may ask root, TLD and authoritative name servers, and finally returns the IP address.
- **Step 4 - TCP connection:** The browser opens a connection to the IP using the TCP three-way handshake (SYN, SYN-ACK, ACK). The data passes through the router, ISP and many routers on the internet (routing).
- **Step 5 - TLS handshake:** For HTTPS the server sends its certificate, keys are exchanged and an encrypted channel is set up.
- **Step 6 - HTTP request:** The browser sends a request such as GET /index.html.
- **Step 7 - Server side:** A CDN, firewall or load balancer may receive it first, pass it to a web server, which may call an application server and database.
- **Step 8 - Response:** The server returns HTML, CSS, JavaScript and images with a status code such as 200 OK.
- **Step 9 - Rendering:** The browser builds the page and loads extra resources, which may cause more requests.

Where is the OSI model here? Steps 6 and 8 are the *application* layer, step 5 is the *presentation/session* layer, step 4 is the *transport* and *network* layers, and the physical cables or Wi-Fi are the *data link* and *physical* layers. We will draw the complete diagram in PDF 3.

9. Revision Summary

- Cyber security is a **protection layer for digital assets** (physical, digital and software-based).
- Information security covers information in all forms and includes **GRC**; hacking is the act of exploiting weaknesses, ethical when authorized.
- There are **14 types** of cyber security: physical, perimeter, application, IAM, cloud, data, e-mail, AI, OS, hardware, firmware, embedded, container, network.
- **OT-ICS security** protects industrial systems (PLC, SCADA); availability and safety are the top priority.
- **Red = offensive, Blue = defensive, Purple = collaboration.**
- The 11 prerequisites: OS, networking, scripting/programming, virus and antivirus, basic security (CIA), database, DNS, certificates, CDN, web server, load balancer and auto scaling.

- **CIA:** Confidentiality, Integrity, Availability.

10. Practice Questions (with short answers)

Q1. What is cyber security?

Answer: Protection of digital assets such as computers, networks, applications and data from attack, damage or unauthorized access.

Q2. Difference between a virus and a worm?

Answer: A virus needs a host file and a human action to spread; a worm spreads by itself over the network.

Q3. Name any 8 types of cyber security.

Answer: Physical, perimeter, application, IAM, cloud, data, e-mail, network (also OS, hardware, firmware, embedded, container, AI).

Q4. What does GRC stand for?

Answer: Governance, Risk and Compliance.

Q5. What is OT and why is its security different?

Answer: Operational Technology runs physical processes; the priority is availability and safety, and systems are old and hard to patch.

Q6. Explain Red, Blue and Purple team.

Answer: Red attacks (offensive), Blue defends, Purple makes both collaborate to improve detection and defense.

Q7. Expand CIA and give one control for each.

Answer: Confidentiality (encryption), Integrity (hashing), Availability (backups and load balancing).

Q8. Why is a CDN used?

Answer: To serve content from nearby servers for speed and to protect against DDoS.

Q9. What is least privilege?

Answer: Giving users only the minimum access they need to do their work.

Q10. What is the difference between a vulnerability, a threat and a risk?

Answer: Vulnerability is a weakness, threat is something that can exploit it, risk is the likelihood and impact of that happening.

End of PDF 1. Reply "next" to get PDF 2: Operating Systems, Types of Servers, Cookies/Sessions/Cache/Buffer, OS Overview and the Boot Process.